Gamifying Occupational Therapy for Prosthetic Training

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Research Experiences For Undergraduates

Introduction

- Lack of functional transfer in training is critical factor towards stagnant 50% prosthesis rejection rate in children[1]
- Manual dexterity assessments evaluate hand functionality[2, 3]
  - **Box and Block Test (BBT)**: records number of wooden cubes transferred across a partition in 60 seconds[2, 4]
- Game-based approaches to prosthesis training has demonstrated advantages in engagement and accessibility[5]
- Virtual Reality (VR) games have added benefit of increased levels of presence, realism and immersion, positively contributing to task execution time and perception of self[6]
  - Previous and ongoing studies have established the feasibility of the BBT in VR[7]

Objective:

- **Gamify the BBT in VR**
- **Evaluate how external parameters affect dexterity**
- **Assess usability and user satisfaction of VR environment**

Methodology

Prototype created in Unity3D™ and deployed on a Meta Quest 3™ headset and controllers

Participant inclusion criteria:

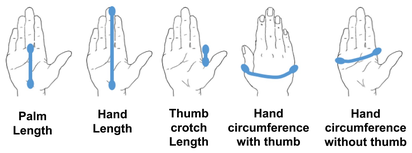
- 18-64 years old
- Full upper extremity control
- No VR-induced sickness



Dynamometer tool used to assess grip strength

Procedure:

1. Gameplay: 3 rounds per arm of the BBT-inspired VR game
2. Measurements: Grip strength and anthropometric dimensions
3. Surveys: Edinburgh Handedness Inventory[8], System Usability Scale (SUS), and Game User Experience Satisfaction Scale (GUESS)



Hand dimensions

Features

Game Flow:

Upon start, idle swimmers spawn

↓

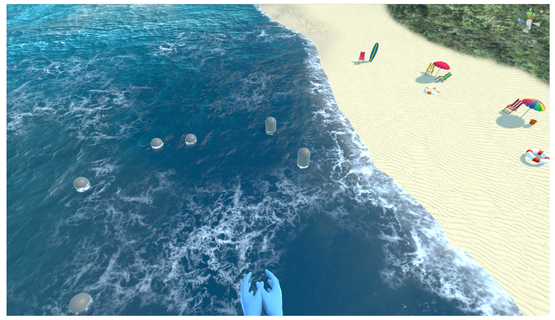
At random, swimmers become distressed

↓

Users have 10 seconds to safely rescue them

↓

Users rescue as many as possible in 1 min.



Screen capture of the VR environment

Scoring:

Rescue Time (sec)	Points
≤ 5	15
6 -10	10
> 10	0

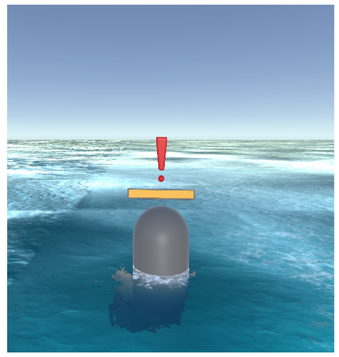
Table 1. Points earned corresponding to time taken to rescue

System Feedback:

- **Swimmer animations**
- **Splash and rescue sound/particle effects**
- **Ambient beach audio**
- **Visual displays of time, score, and total persons rescued**

Design Considerations

- Design components prioritized training transfer but with interactivity and randomization (not “training to the test”)
  - Left and right-handed setup
  - Time-based rescue
  - Beach-themed design
- Grounded in a non-violent game design approach
- Designed for use in standing position with minimal physical movement for ease with grasp and release action
- Controller models were replaced with hand models to enhance immersion and realism



Person placeholder with rescue timer bar and exclamation point when in distressed state

Future Work

Short-term:

- Continue iterative design of prototype:
  - Implement further visual and auditory features
  - Refine hand physics and grasp mechanics
  - Incorporate haptic feedback to simulate grabbing sensation
- Conduct proposed experimental study
  - Apply user feedback in order to introduce the game to patients with upper limb difference



Patient with congenital limb difference wearing her Limbitless Solutions prosthetic

Long-term:

- Incorporate EMG flex controller input and render a simulated prosthetic arm
- Evaluate performance disparities between the VR BBT simulation and game
- Study the long-term effects of incorporating serious VR games into prosthetic training

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